DXMapper Suite — Owner's Manual

FM Tuner · RDS Monitor · DX Mapper

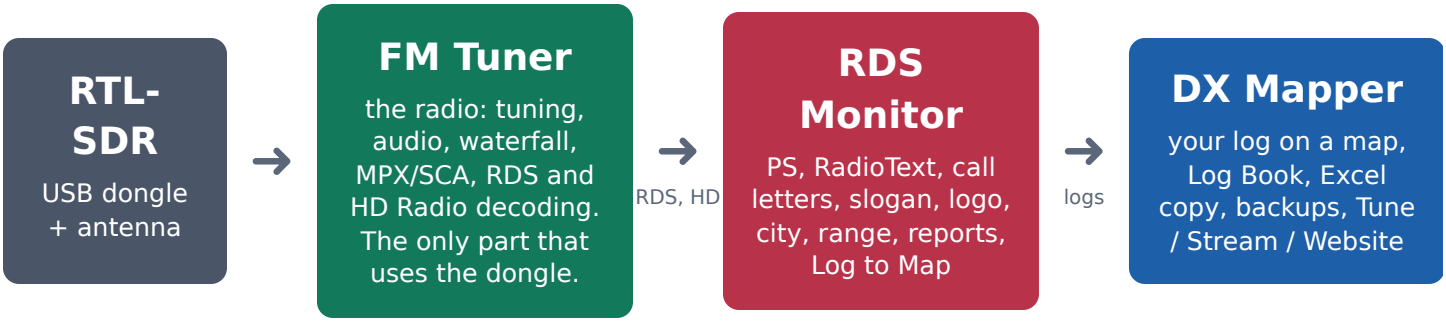
For DXing and for everyday listening.

Contents

1. The three parts
2. Installing and updating
3. Starting and arranging the windows
4. FM Tuner
4A. Remote – listen on your phone, tablet or another PC
5. RDS Monitor
6. DX Mapper
7. Your data – the settings folder
8. Troubleshooting
9. Glossary
10. For experimenting
11. Credits

1. The three parts

DXMapper Suite is one program in three parts that run side by side on the same PC and talk to each other. Each part has one job, and each can also run on its own.



↶ Tune buttons in the RDS Monitor and DX Mapper tune the FM Tuner

Who does what

Part	Job	You use it to
FM Tuner	Owns the RTL-SDR dongle. Turns the radio signal into sound and data.	Tune, seek, save presets, listen, watch the spectrum, decode HD Radio, record MPX.
RDS Monitor	Reads the RDS data the tuner decodes and shows who the station is.	Identify a station (call letters, city, distance, logo), read RadioText and RT+, log a catch.
DX Mapper	Keeps every catch you ever logged, forever, and puts it on a map.	See your log on a map, read it as a Log Book, import old logs, back up, tune or stream a station.

DX and everyday listening

Distance decides a station's range everywhere in the program:

Range	Distance from you	What it means
Local	0–40 miles	Comes in every day
Extended local	41–80 miles	Comes in sometimes, with conditions
DX	81 miles and up	The real catch

DX Mapper's station list starts with your Locals and Extended locals in frequency order, like a preset list, so the same program works for chasing DX and for just listening to locals. The miles can be changed under DX Mapper › **Home…**; the RDS Monitor uses the same miles.

2. Installing and updating

First time

1. **Install Python 3.10 or newer, 64-bit,** from `python.org/downloads/windows`. In the installer tick *"Add python.exe to PATH"* and keep *"tcl/tk and IDLE"* ticked (the windows need it). If you are offered Python 3.14 and the install step fails, use Python 3.12 instead.

2. **Extract the ZIP** into `C:\` (or wherever you keep programs). It holds one folder, **DXMapper Suite**, so you get `C:\DXMapper Suite`. In Windows "Extract All", change the path to `C:\` so you don't get the folder twice. DXMapper Suite is one program in one folder: the FM Tuner, RDS Monitor and DX Mapper are its three windows.

3. **Double-click "Install All.bat".** It installs what all three parts need and tells you if anything is missing.

4. **Dongle driver (once):** plug in the dongle, run **Zadig** (`zadig.akeo.ie`), choose *Options › List All Devices*, pick the dongle ("Bulk-In, Interface (Interface 0)" or "RTL2838UHIDIR"), choose **WinUSB** and click *Replace Driver*. Repeat for each USB port you use.

5. **rtlsdr.dll (once):** download the RTL-SDR Blog `Release.zip`, open its **x64** folder and copy `rtlsdr.dll` and `libusb-1.0.dll` into the main folder (next to Start Radio). This copy also supports Bias-T.

```
C:\DXMapper Suite\
  Start Radio.pyw      every day: opens all three
  FM_Tuner.py          quick launch: just the radio
  RDS Monitor.py       quick launch: just the RDS display
  DX Mapper.py         quick launch: just the map log
  settings\            ALL your data, for all three (never in the ZIP)
  rtlsdr.dll, libusb-1.0.dll you add these (step 5)
  INSTALL.txt          detailed install steps
  docs\                the three programs' READMEs
  QUICK_START.pdf      the short guide
  OWNERS_MANUAL.pdf    this manual
  DEPENDENCIES.txt     every part the programs need, and where it comes from
  Install All.bat      first time, and after a Python update
  Arrange Windows.bat  choose where each window goes
```

DEPENDENCIES.txt lists every part the three programs use: what is already in the ZIP, what Install All adds, and what you add yourself (rtlsdr.dll, Zadig, PothosSDR for HackRF). The HD decoder is optional and not included.

Updating to a new copy

Extract the new ZIP to the same place (`C:\`), over the old copy, exactly as above. **Coming from a copy without the DXMapper Suite folder** (for example `C:\Radio`)? Rename that folder to **DXMapper Suite** once before you extract, so your settings stay with it. The ZIPs never contain the **settings** folder, so your presets, layout, stations file, logos, locations, log and backups stay exactly as they were. You never need to redo anything after an update.

WHATS_NEW.txt, next to Start Radio, lists every update by date, newest first, so you can see what changed since your last copy. This manual and the Quick Start are updated with each new copy.

Free public copy: the copy for sharing (*DXMapper Suite v<release>.zip*, the number is the release date, e.g. v2026.10.12) has all the same features and none of your stations, logs or settings. It adds **EULA.txt** (the free license: use and share it free, don't sell it) and **THIRD_PARTY_NOTICES.txt** (the outside parts it uses and their licenses). Give people that copy, never your own folders.

Coming from the old three-folder copy? Rename the folder that holds them to **DXMapper Suite** and extract the new ZIP over it. The first time you start it, your settings from the old `FM-Tuner`, `RDS-Monitor` and `DX-Mapper` folders, and rtlsdr.dll, are **copied** into the one `settings` folder. The old folders are left untouched, so you can go back to the old copy; delete them once everything works.

> If Windows asks whether to replace files, say **Yes**. Only program files are replaced.

The HD Radio decoder

HD Radio decoding uses the free **nrsc5** decoder, which is not part of the public download. If you have the decoder add-on, install it once from the FM Tuner (see *4.6 HD Radio*). It is stored in `settings\hd`, so updates keep it too.

Running from a thumb drive (travel laptop)

Everything the programs need sits in one folder, so the whole radio can live on a USB thumb drive. After a laptop has been set up once, all you carry is the **thumb drive, the RTL-SDR and an antenna**. Your presets, Log Book, logos and settings travel with the drive.

Put the radio on the drive (once):

1. Extract the ZIP to the drive (for example `E:\`, giving `E:\DXMapper Suite`), the same way as above.
2. To take your existing log and presets along, copy the **settings** folder from your PC into the same folder on the drive.
3. Copy `rtlsdr.dll` and `libusb-1.0.dll` into the main folder on the drive. They travel with it.

Set up each laptop (once per laptop):

1. Install Python 3.10 or newer, 64-bit, with *"Add python.exe to PATH"* ticked (step 1 above).
2. Plug in the drive and double-click **"Install All.bat"** on the drive. It installs what the programs need onto that laptop.
3. Plug in the dongle and run **Zadig** (step 4 above). Repeat for each USB port you use on that laptop.

Every trip: plug in the drive, the dongle and the antenna, then double-click **"Start Radio.pyw"** on the drive. In the RDS Monitor use *Tools › Set Away Location...* (or *GPS Location...* with a USB GPS dongle) and *Use Away location* so the map and distances follow you.

> - **The drive letter can change** (E: on one laptop, F: on another). The programs find their settings and each other wherever the drive is.

- **Close all three programs before you unplug the drive**, then use Windows' "Safely remove". Pulling it out while DX Mapper is open can damage the log.
- **One log, one place.** A copy on your PC and the copy on the drive keep separate logs. Either run from the drive everywhere, or copy the settings folder back to your PC when you get home.
- DX Mapper's optional extra backup folder (for example Google Drive) is a full path, so on a laptop without that folder only the backups in `settings\Backups` on the drive are made.

3. Starting and arranging the windows

Every day

Double-click **Start Radio.pyw**. It opens the FM Tuner, then the RDS Monitor, then DX Mapper, with no black console windows. The RDS Monitor connects to the tuner by itself, and DX Mapper listens for the monitor. Any part can be closed and opened again at any time; the others reconnect on their own.

Where the windows go

Once the windows open, Start Radio puts them in place for the number of screens it finds:

Screens	Layout
1	FM Tuner on the left third; RDS Monitor top right; DX Mapper bottom right.
2	FM Tuner on display 1. RDS Monitor above DX Mapper on the **left half** of display 2, so the right half stays free (for research, YouTube…).
3 or more	One program per screen.

Run **Arrange Windows.bat** to choose which display each program goes on, whether it takes the whole screen, the left half or the right half, and whether Start Radio arranges them every time. The choice is saved in the settings folder next to it. You can still move and resize the windows by hand afterwards.

Remembers your layout: while the programs run, Start Radio notes where each window is (and whether it's maximised) and puts them back there next time. It remembers one layout per screen setup, so unplugging the TV falls back to the normal arrangement and plugging it back in brings your TV layout back. A window that would land off-screen gets the normal place. To go back to the normal arrangement, press **Arrange now** in Arrange Windows; untick *Put windows back where I last had them* to turn it off.

On a 4K TV or a QHD monitor, the FM Tuner's *View › Size* (100–200 %) makes its text larger. If HDR is on in Windows and the windows look dim or washed out, adjust *Settings › Display › HDR › SDR content brightness*.

4. FM Tuner



FM Tuner window (opened without a dongle, so no signal or RDS is showing).

The window is made of panels: Tuning, Presets, Signal, RF spectrum + MPX, Gain, Antenna power, Audio, Live RDS, Localhost services and Log. The **View** menu turns panels on and off.

4.1 Tuning

- **Frequency:** click the frequency, type a new one (e.g. `101.1`) and press Enter. The US FM band on your region's band and step is accepted (Americas: 87.7–108.1 MHz in 0.2 MHz steps).
- **−0.2** / **+0.2** step one channel; they wrap around at the band edges.
- **Seek Down** / **Seek Up** move off the current station and stop on the next channel that stands clearly above the band noise (mono stations included). If nothing is found in a full lap it goes back where it started.
- **HD Seek Down** / **HD Seek Up** do the same, but on every station they hear they wait up to 8 seconds for the HD decoder to lock, and only stop where it does. They need the HD decoder add-on and are **hidden until it is added** (Audio panel › Add HD decoder…); the remote page hides them too. **Stop** ends any seek.
- **Outside the Americas** (Region set to Europe, Japan or OIRT) there is no HD Radio on the air, so **every HD control is hidden**, including Add HD decoder…. Set Region back to Americas to see them again.
- **Stop** / **Start** release and take the dongle again.

4.2 Presets

- **+ Save current** stores the station you are on, named after its call letters or PS when RDS has decoded one.

- One click on a preset tunes to it. A preset on the frequency you are tuned to lights up blue.
- **Right-click** a preset to select it (green), then use **Rename**, **Delete** or the arrows to move it.
- Presets are kept in `settings\tuner_presets.json` and survive updates.

4.3 Signal and gain

- The gain starts on **manual, 0.0 dB**. Raise it for weak stations; lower it if strong locals overload the dongle. Auto gain is also available.
- **Remember gain per frequency** (tick box right under Manual/Automatic, off to start) and the **Save gain** button next to it: while it's on and gain is on Manual, set the gain you want and click **Save gain**. That gain is kept for the frequency you're on (the Log shows *[gain] 99.7 MHz: saved 20.0 dB*). **Moving the slider, + / − or the mouse wheel never saves on its own.** The saved gain comes back each time you tune there again (by hand, preset, seek or the remote) and when you press Start. Auto gain saves and recalls nothing. Saved gains are kept in `settings\gain_memory.json`.
  A frequency only has a saved gain after you've clicked Save gain there once; until then it keeps whatever gain you had. Clicking Save gain with the box unticked, or on Auto, just pops up what to do first. Each time a saved gain comes back, the Log shows a line like *[gain] 99.7 MHz: remembered 20.0 dB*.
- The mouse wheel moves the gain and volume sliders (gain one step at a time, volume 0.1 at a time).
- Point at the big frequency in the TUNING panel and roll the mouse wheel to tune: one channel per notch at your region's step (Americas: 88.1, 88.3…).
- **Region** menu: the first time a new install starts, the tuner asks for your region. **Americas** 87.7–108.1 MHz in 0.2 MHz steps on odd frequencies; **Europe / most of the world** 87.5–108.0 MHz in 0.1 MHz steps; **Japan** 76.0–95.0 MHz in 0.1 MHz steps; **Eastern Europe OIRT** 65.90–74.00 MHz in 0.03 MHz steps. The step buttons, mouse wheel, arrow keys, seek, the frequency box and the phone remote all follow it. An existing install stays on Americas, and presets from another band are kept.
- The **ADC readout** beside the signal bar says *ADC ok*, *ADC near clip* or **OVERLOAD**. Once the dongle clips, no filter can undo it: lower the gain, lower the LNA gain, or add an FM notch / attenuator.

4.4 Audio

- **Audio** on/off, **Stereo**/mono and **Volume**.
- **Output device:** pick your speakers. If you hear nothing, make sure it is not an HDMI / “Display Audio” device, then press Refresh.
- **Listen to:** Main programme, or an analog **SCA** subcarrier at 67 or 92 kHz (reading services, background music). Beside it, *SCA PRESENT* (green) or *no SCA carrier* (red) says whether the station sends one. Most US stations send none.

Radio: RTL-SDR or HackRF

> **HackRF and SoapySDR support is EXPERIMENTAL until it is fixed.** It may not work right; an RTL-SDR dongle is the supported radio.

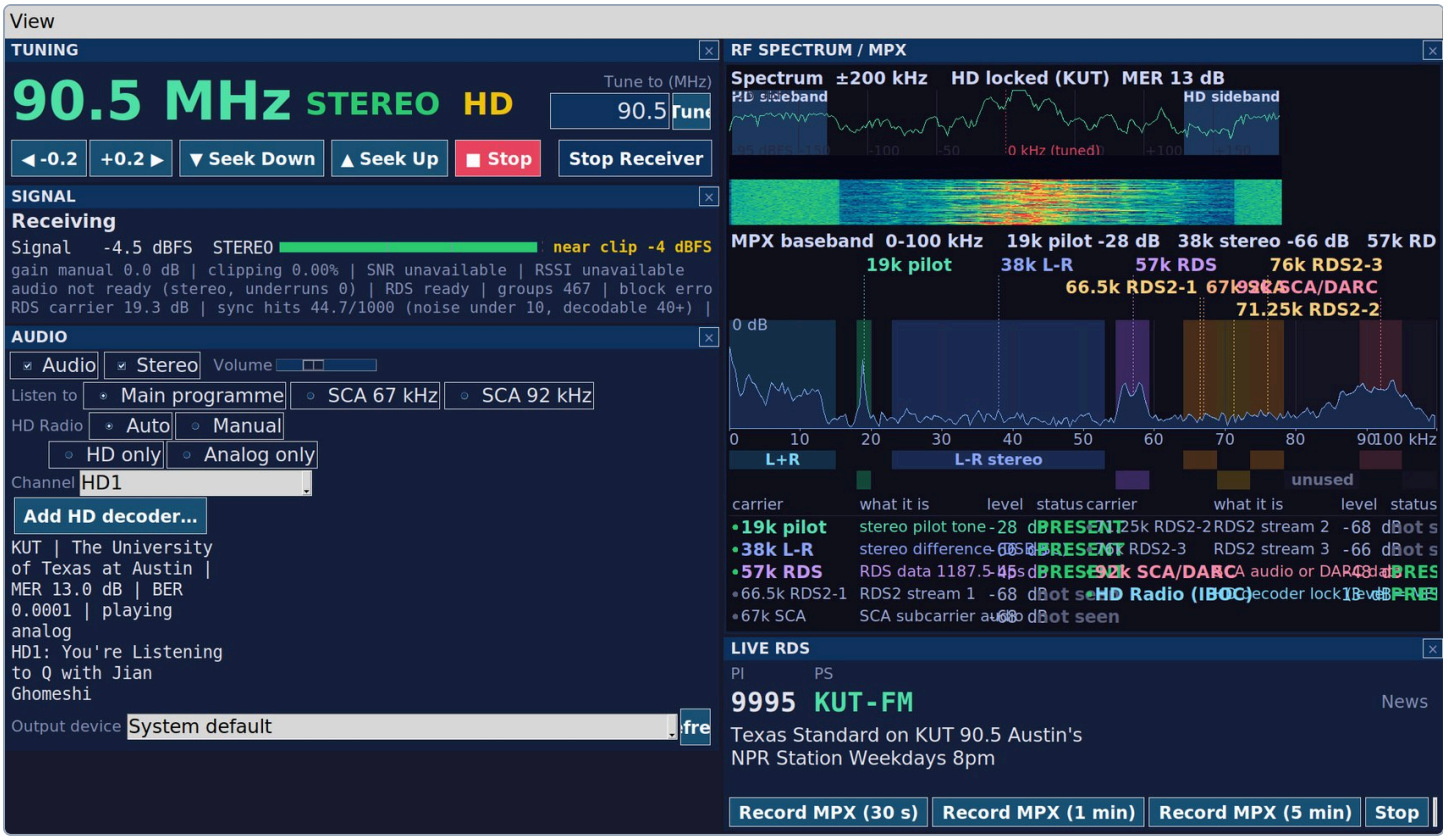
The **Radio** list at the top of the Gain panel picks the radio (the Gain panel starts hidden; turn it on with *View › Gain*). It shows your RTL-SDR dongles and, once the free PothosSDR (SoapySDR) package is installed, a HackRF too (INSTALL.txt, FM Tuner step 4a). Press **Find** after plugging one in. If no

HackRF is listed, the line under the list says why: SoapySDR isn't installed, or it is but can't see the HackRF (check the cable and driver). The tuner remembers your choice. A HackRF runs at 1.92 MHz and is sliced down to about 1 MHz, the same as the RTL-SDR, so FM, RDS and HD work the same. Its gain goes 0–102 dB with the 14 dB RF amp left off (it has no automatic gain, so **Auto** sets a fixed 48 dB; use Manual to change it), and Bias-T isn't switched from the tuner on a HackRF. The Log Book's Source column shows which radio made each catch.

4.5 Antenna power (Bias-T)

The **Bias-T** box puts 4.5 V on the antenna socket for a powered LNA or active antenna. It is always off at start-up, asks before switching on and switches off when the dongle is released. **Only use it with a powered LNA or active antenna connected**: a DC-shorted antenna or splitter can damage the dongle.

4.6 HD Radio



FM Tuner locked to an HD station (nrsc5's recorded KUT sample, not off the air): orange HD badge, Channel list and HD details in the Audio panel.

Installing the decoder (once): in the Audio panel click **Add HD decoder…** and pick the decoder ZIP (or `libnrsc5.dll`). It is copied into `settings\hd`.

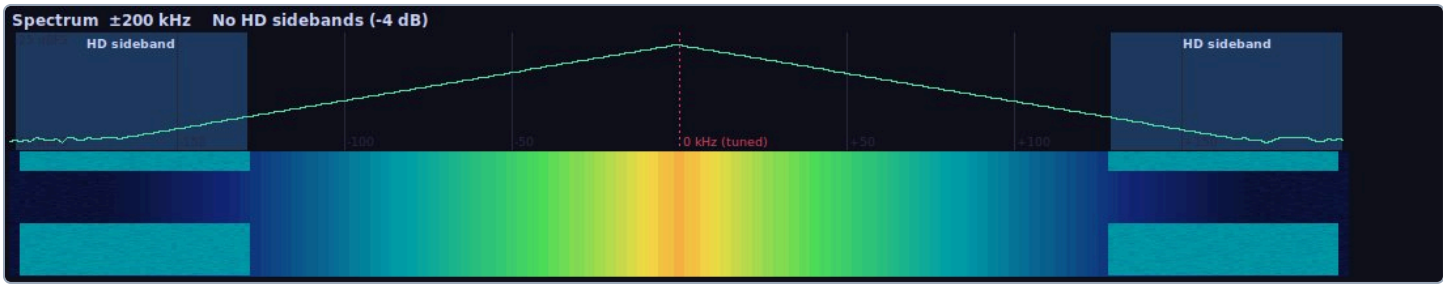
Audio panel shows	Meaning
HD decoder not installed – analog FM works as normal	The decoder files are not in settings\hd yet.
HD: no lock on this frequency	The decoder is installed; this station isn't giving it HD (no HD, or too weak).

HD locked (call) | MER nn dB good	Decoding HD. MER is the digital signal quality; higher is better. It is colour-coded: green *good* (10 dB or more, solid HD), yellow *fair* (6–10 dB, some dropouts), red *weak* (under 6 dB, mostly analog).

- **Auto:** switches to HD after about 3 seconds of steady lock, and fades back to analog when HD drops.
- **Manual:** plays the channel you pick and never switches by itself.
- **HD only:** digital audio only; silent without HD. **Analog only:** never plays HD audio, but still reads the HD station info.
- **Channel:** Analog plus only the HD channels the decoder has actually found here (HD1, HD2 “The Loop”…). It empties on every retune.
- **HD channel buttons:** the same choice as one-click buttons in the Tuning panel, under the tune and seek buttons. Only the station's own channels appear (Analog, HD1, HD2…), labelled with the decoded name such as *HD2 · The Loop*; the one playing is green. The row only shows on a station with HD.
- **Analog always plays live**, so manual tuning gives instant sound. Like a table-top HD radio, the station delays its own analog to line up with HD.

Getting a lock: HD needs a much stronger signal than analog. On the waterfall, HD shows as flat shelves 130–200 kHz either side of the station. If you can't see those shelves above the noise, it will not lock. Aim the antenna, try a little less gain if strong locals are nearby, and try again when conditions are good.

4.7 RF spectrum and waterfall

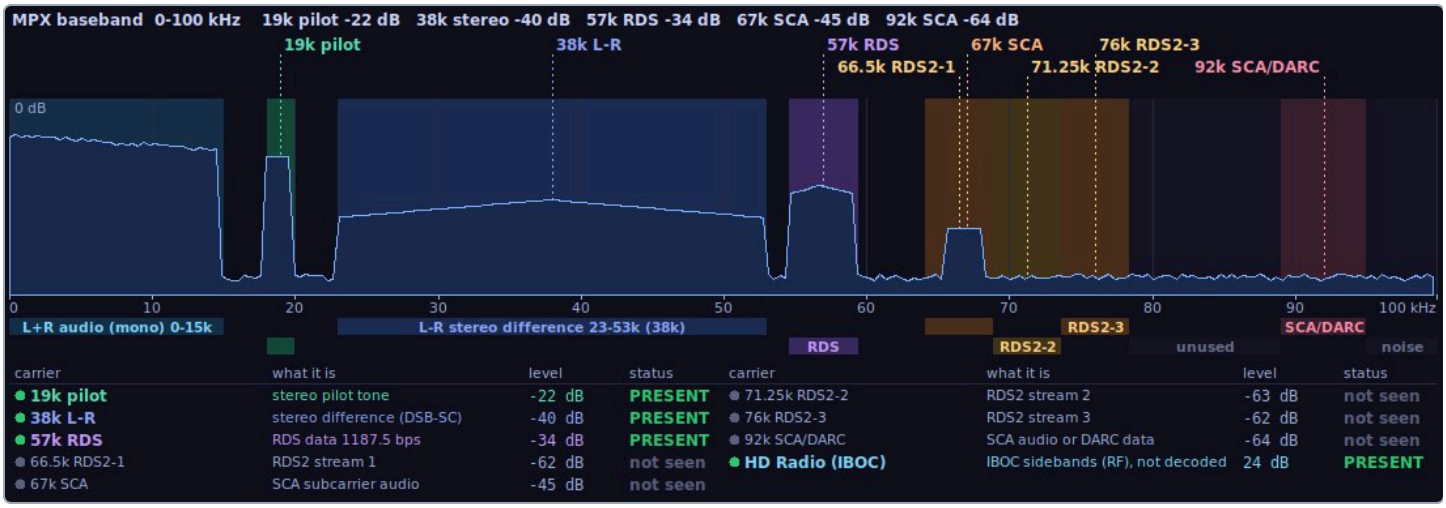


RF spectrum and waterfall, ±200 kHz around the tuned station. The shaded bands are where HD Radio sidebands live. Demo picture.

The waterfall scrolls down: the newest line is at the top.

The header gives the HD decoder's verdict: **HD locked**, **No HD lock**, **HD lost** or **HD decoder not installed**. Only a real decoder lock counts as HD.

4.8 MPX baseband – what the station is sending



MPX baseband, 0–100 kHz, with every carrier labelled and marked PRESENT or not seen. Demo picture.

Marker	What it is
L+R audio 0–15k	Mono programme audio
19k pilot	Present = the station is transmitting stereo
38k L−R	Stereo difference signal
57k RDS	Classic RDS; a visible bump means RDS is on the air
66.5k / 71.25k / 76k	RDS2 subcarriers (very rare in the US)
67k SCA, 92k SCA/DARC	Analog SCA subcarriers or DARC data
HD Radio	PRESENT only while the HD decoder is locked

The MPX display tells “no RDS transmitted” apart from “RDS too weak to decode”.

4.9 Live RDS

- PI and PS as decoded, plus RadioText.
- **RDS carrier** and **sync hits**: under 10 sync hits is noise; a decodable station reads about 40 or better.
- **RDS2 streams** (off by default) decodes the RDS2 subcarriers. Experimental: almost no US station sends RDS2 yet.

4.10 Record MPX

Record MPX (30 s / 1 min / 5 min) saves the composite signal as a WAV file that RDS Spy and SDR# can open (1 minute is about 29 MB). A timer shows while recording; **Stop** ends early and keeps the file.

4.11 DX / interference filter

View › DX / interference filter helps with a weak station 200 kHz from a strong local.

Bandwidth	Use it for

56–80 kHz	Deep DX right beside a strong local. Thin mono audio; too narrow for RDS or stereo.
110–130 kHz	Everyday DXing: clean mono, still decodes RDS.
150–180 kHz	A clear channel: good mono or light stereo, fast RDS.
200–230 kHz	Normal broadcast width (default). Needed to see HD sidebands.

- **Protect from:** Off, Auto, Low side (−200k) or High side (+200k) cuts the passband short on the side of the strong neighbour only. Auto picks the side that is 6 dB or more stronger.
- **Canceller** subtracts what still leaks through from the protected side.
- One dongle can't separate two stations on the *same* frequency.

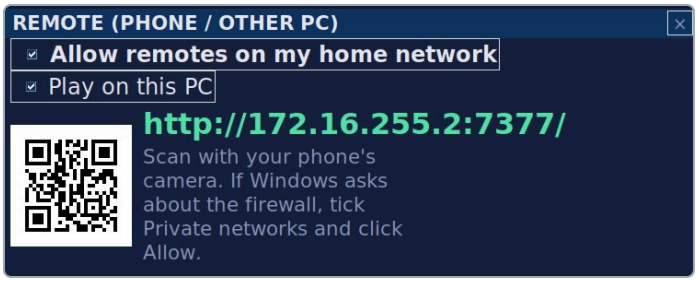
4.12 View menu and layout

- **Tuner only** shows just Tuning, Presets and Live RDS; **Show all panels** shows everything.
- Drag a panel by its title bar into either column (a green line shows where it lands); × hides it; **Default columns** puts them back.
- **Always on top**, **Size** (100–200 %, after a restart) and **Layout presets** (save the arrangement under a name like “TV 4K” and recall it later).
- The layout is saved in `settings\tuner_layout.json`.

4A. Remote – listen on your phone, tablet or another PC

The FM Tuner can play in every room of the house: any phone or tablet (iPhone, iPad or Android) or another PC or Mac on your home Wi-Fi opens a web page from the tuner that shows the station and plays its audio. It is web based, so it works in Safari, Chrome or any other browser, and nothing has to be installed on the phone or tablet. There is only one dongle, so **every remote hears the same station**; if one remote changes the station, it changes for all of them.

Switching it on



The REMOTE panel in the FM Tuner with the remote switched on: the address, the QR code and Play on this PC. Test setup.

1. In the FM Tuner, find the **REMOTE (PHONE / TABLET / PC)** panel (*View* menu if it is hidden) and tick **Allow remotes on my home network**.
2. The panel shows the address, for example `http://192.168.1.20:7377/`, and a **QR code**.
3. On the phone, open the camera, point it at the QR code and tap the link. On another PC or Mac, type the address into Chrome, Edge or Firefox. The phone or PC must be on the same home network as the tuner PC.
4. The first time, Windows may ask whether to allow Python on networks: tick **Private networks** and click **Allow**. If you clicked Cancel by mistake, see Troubleshooting.
5. On the remote page, press **Listen on this device**. Press it again (**Stop listening**) to stop.

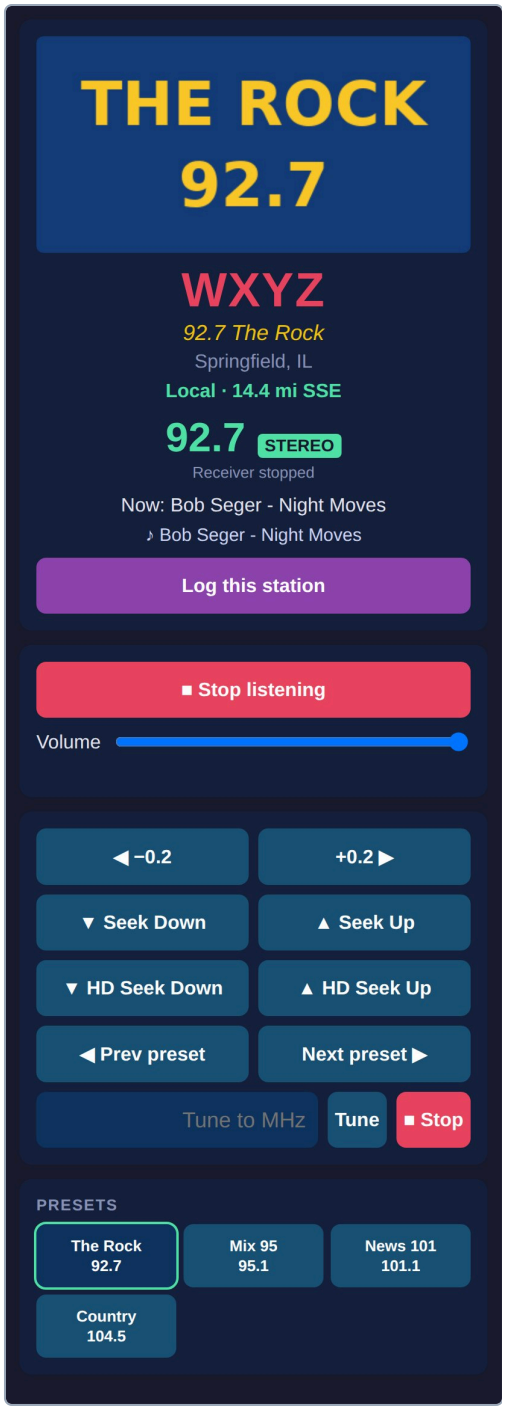
The tick is remembered, so the remote comes back on the next time the tuner starts. Untick it to close the page to everyone.

Play on this PC

While the remote is on, a second tick box appears: **Play on this PC**. Untick it to silence the tuner PC's own speakers while the phones and other PCs keep playing – handy when the tuner PC is in another room. Switching the remote off turns the PC's sound back on by itself, so the tuner can't be left silent.

The remote page

The remote page on a phone. Made-up test station and logo; 'Receiver stopped' because no dongle was connected.



- **Top:** the station logo, call letters, slogan, city, and range with miles and direction (Local / Extended local / DX, the same 40/80-mile tiers as DX Mapper).
- **Then:** the frequency with STEREO and HD badges, the signal, RadioText and now playing. When an HD station has HD2, HD3… buttons appear for them.
- **Listen on this device** and a volume slider for that phone or PC only.
- **Buttons:** −0.2 / +0.2, Seek Down / Up, HD Seek Down / Up (only with the HD decoder, Americas region), Prev / Next preset, a Tune to MHz box and Stop.
- **Presets:** one button per preset; the one on the air is outlined in green.
- **Log this station** logs the catch to DX Mapper, like the RDS Monitor's Log to Map – for remote DXing from another room.
- On a phone, the lock-screen previous / next track buttons go to the previous / next preset.

Logo, call letters, slogan, city and range come from the RDS Monitor and DX Mapper, so run them as usual (Start Radio does). Without them, the page shows the tuner's own PS and RadioText.

Good to know

- Remote audio runs about 1–3 seconds behind the tuner PC; that is normal for network audio.
- The page only answers devices on your home network. It can tune, seek and change presets, but it can't change gain, delete presets or shut the tuner down.
- The phone's screen can turn off while listening on most phones; if the audio stops when it locks, keep the browser tab in front.

Arrow-key hotkeys (FM Tuner window)

Key	Does

↑ Up arrow	Tune up one channel (region step)
↓ Down arrow	Tune down one channel (region step)
← Left arrow	Previous preset; from the first preset it wraps around to the last
→ Right arrow	Next preset; from the last preset it wraps around to the first

The keys work anywhere in the tuner window except while you're typing in a box. *View › Arrow keys* switches them off.

Signal units: *View › Signal units* shows the signal as S-units (S1–S9, then S9+10…, the default) or dB (dBFS, for technical users). The RDS Monitor, the remote page and DX Mapper (Station Details, the log lists and the Excel copy) follow the tuner's choice. The dongle only measures dBFS, so S-units are relative (S1 at −70 dBFS, 6 dB per S-unit, S9 at −22 dBFS), not calibrated like a ham radio's meter.

5. RDS Monitor



RDS Monitor following the tuner (KUT recording). Logo, call letters, slogan and city on the left; PI, PTY, PS, now playing and RadioText in the middle; RT+ and the HD RADIO panel on the right; signal quality along the bottom.

5.1 Where the RDS comes from (Input menu)

Input	Source
FM Tuner (TCP)	The FM Tuner. Default; it keeps retrying until the tuner is running.
Spy File (RDS Spy)	A .spy file, read as it grows.
RDS Input	Any other RDS source sending ASCII G groups, e.g. SDR#'s RDS output on port 23.

5.2 The tuner bar

With the FM Tuner input, a bar under the header shows the tuned frequency in large text, the signal, stereo, HD and gain, and has Tune, −0.2, +0.2, Seek, Stop and Start buttons plus **Audio** and **Bias-T** boxes. Roll the mouse wheel over the Tuned frequency to tune one channel per notch (the tuner's region step). It follows the tuner whichever window you tune from. **Dock** snaps the monitor beside or below the tuner window and follows it.

5.3 Who is this station?

- **Call letters** come from the PI code, using **DX Mapper's log** first (every log, so old Historical stations are named too), then the PI-code math.
- **Automatic call letters from the PI code work for US stations only.** Outside the US (Canada, Mexico, Europe…) a PI code doesn't spell out call letters, so the monitor shows *unknown* until you name the station once in DX Mapper (Edit); after that it is shown every time.
- **Slogan** (in italics), **city**, and the **range and distance** with compass bearing, for example *Extended local · 66.2 mi NE (41°)*.
- **Locations** come from the station's Wikipedia page, or the FCC database when Wikipedia has none.
- **Logos** come from the logo folder, then Wikipedia, then the station's Facebook page. *Manual Logo Lookup* lets you paste a Facebook page address if nothing is found.
- **Wrong or unknown call?** Fix it in DX Mapper (Edit info or the Log Book). The monitor reads the log and relabels the station within a second, no restart. (The old Set Callsign and PI Overrides menu items are gone; the log does that job now.)

5.4 Station names come from DX Mapper's database

The monitor gets call letters, city, coordinates and slogan straight from **DX Mapper's log database** (every log, read only). There is no PI override file to edit any more: to fix or add a station, edit it in DX Mapper and the monitor shows it within a second.

Behind the scenes the monitor still keeps a small `settings\stations.txt` with stations it worked out from the PI code before they are in any log. You never need to open it.

5.5 What's on the screen

- **PS**, **Long PS** (when sent), **PTY** / **PTYN**, **RadioText**, clock time, TP/TA and music/speech flags.
- **RT+**: labelled rows (Title, Artist, Album, Now, Next, Host, Website…) for whatever the station tags.
- **Now playing** and temporary **cover art**: while HD is locked, the album art the station sends over HD for the channel you're playing; otherwise looked up online. Never saved.
- **TMC** traffic messages: a ticker scrolling along the bottom of the window, so a pile of RT+ rows can never push it out of sight. It shows only while the station sends TMC; point at it to pause it. *View › Show TMC Traffic Messages* turns it off.
- **Music search buttons** under the title and artist: **YTM**, **SPOT**, **AMZ** and **APL** search the song on YouTube Music, Spotify, Amazon Music or Apple Music in your browser. They light up by themselves in each service's colour while a song is showing (RT+ or RadioText) and go grey for promos, ads and station IDs.
- **HD RADIO** panel (with the HD decoder): lock, MER (colour-coded green/yellow/red, the same as in the FM Tuner) and BER, station name, slogan, HD1–HD8 names and now playing, alerts.
- **Signal quality**: how much of the RDS was decoded, block rate, and decode badges (green = complete, yellow = partly received).

- **SEQ>** line under the badges (green, console style): the last 24 RDS group types in the order they arrived, e.g. 0A 0A 2A 0A 8A. Each station's encoder repeats its own pattern, so it works like a fingerprint. It goes to DX Mapper with every log (Log Book column **Group Sequence**).
- **Lifetime** (unique PI codes in your current DX Mapper log, the first one ticked) and **Session** station counters.

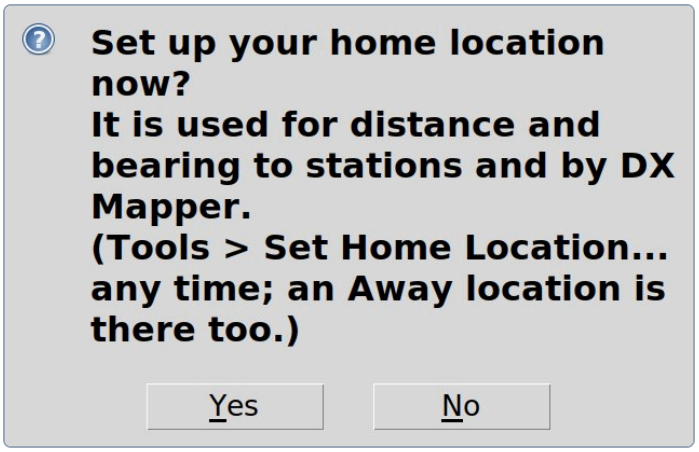
5.6 Logging a catch

- **Log to Map** (under the call letters) or **Ctrl+L** sends the station and its full RDS report to DX Mapper. The button then says *Logged*, or *Map not running*. Pressing it yourself always logs the station on the frequency you're tuned to, even when its PI is already in your log somewhere else; that's your override for the splatter rule (see 6.5).
- *Tools › Auto-log to DX Mapper* (off by default) logs a station once 60 % of its info is decoded. While the station keeps decoding it sends an update every 5 minutes, and sooner when something new decodes (RT+, AFs, a new ODA, Long PS...). Every update fills in the same Log Book entry, so the log picks up new info over time instead of keeping only the first decode.
- *Tools › └ Auto-log HD catches too* is a sub-option of Auto-log (off by default, greyed out while Auto-log is off). When the FM Tuner's HD decoder holds lock for 5 seconds, the station is logged right away with its HD info (call, slogan, HD1–HD8 channels, MER), even before the RDS reaches 60 %. While HD stays locked it updates every 5 minutes. A lost lock starts the 5-second wait over.

5.7 Home and Away

Tools › Set Home Location... (asked on first start) and *Set Away Location (DX on the go)...*. **GPS (optional):** with a USB GPS dongle plugged in, *Tools › GPS Location (USB GPS dongle)...* finds it by itself (or pick its COM port), shows the fix, and has **Set as Home** and **Use as Away location**. Tick **Keep the Away location updated from GPS while traveling** and Away follows you, saved at most once a minute and only after you move half a mile; it starts again by itself next time. No dongle needed: typing Home/Away works as before. GPS uses the pyserial add-on that Install All puts in.

Use Home location / *Use Away location* picks where distances are measured from; DX Mapper follows, and every log remembers where you were.



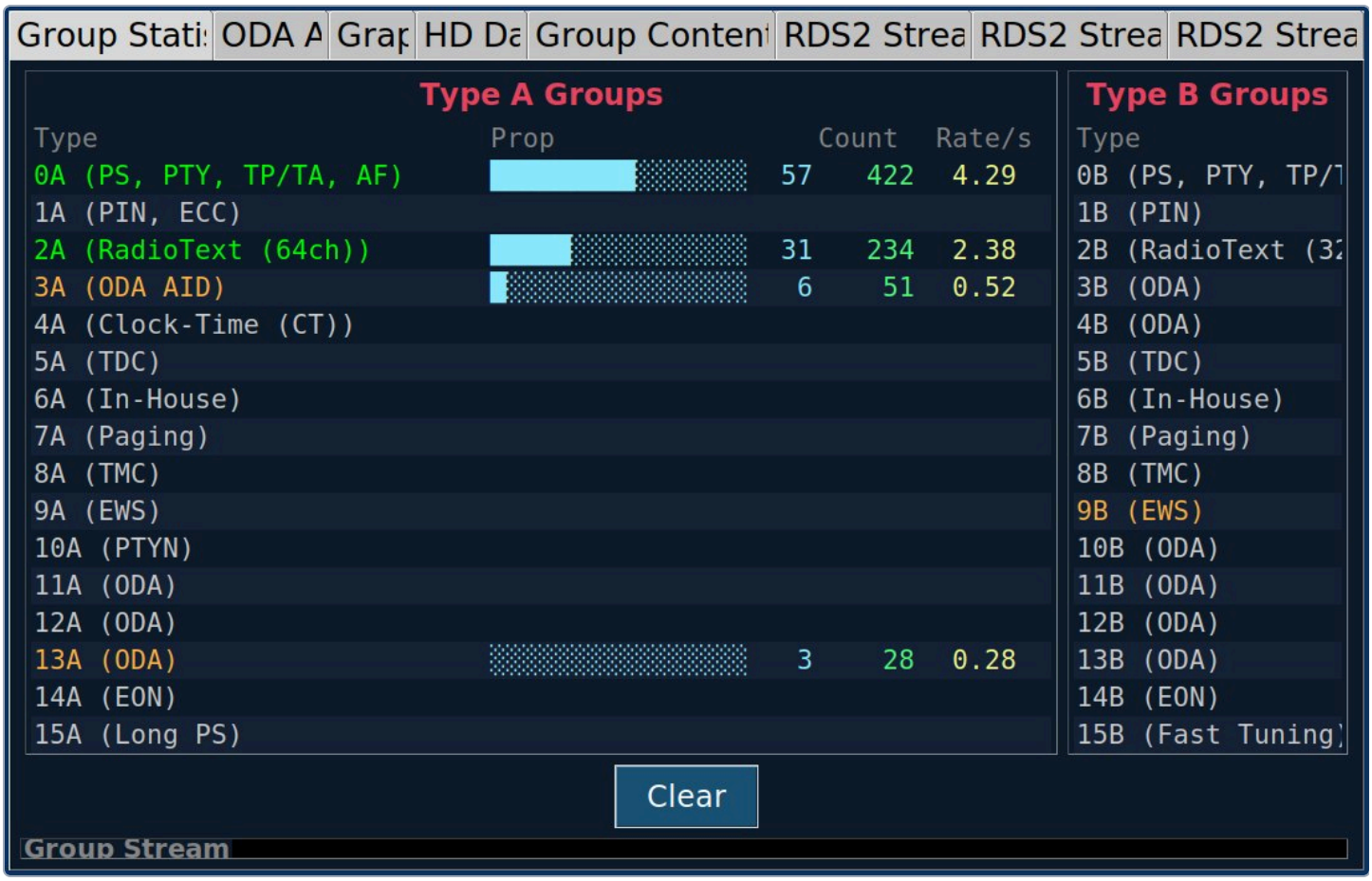
First start asks for your home location.

5.8 Group Analyzer

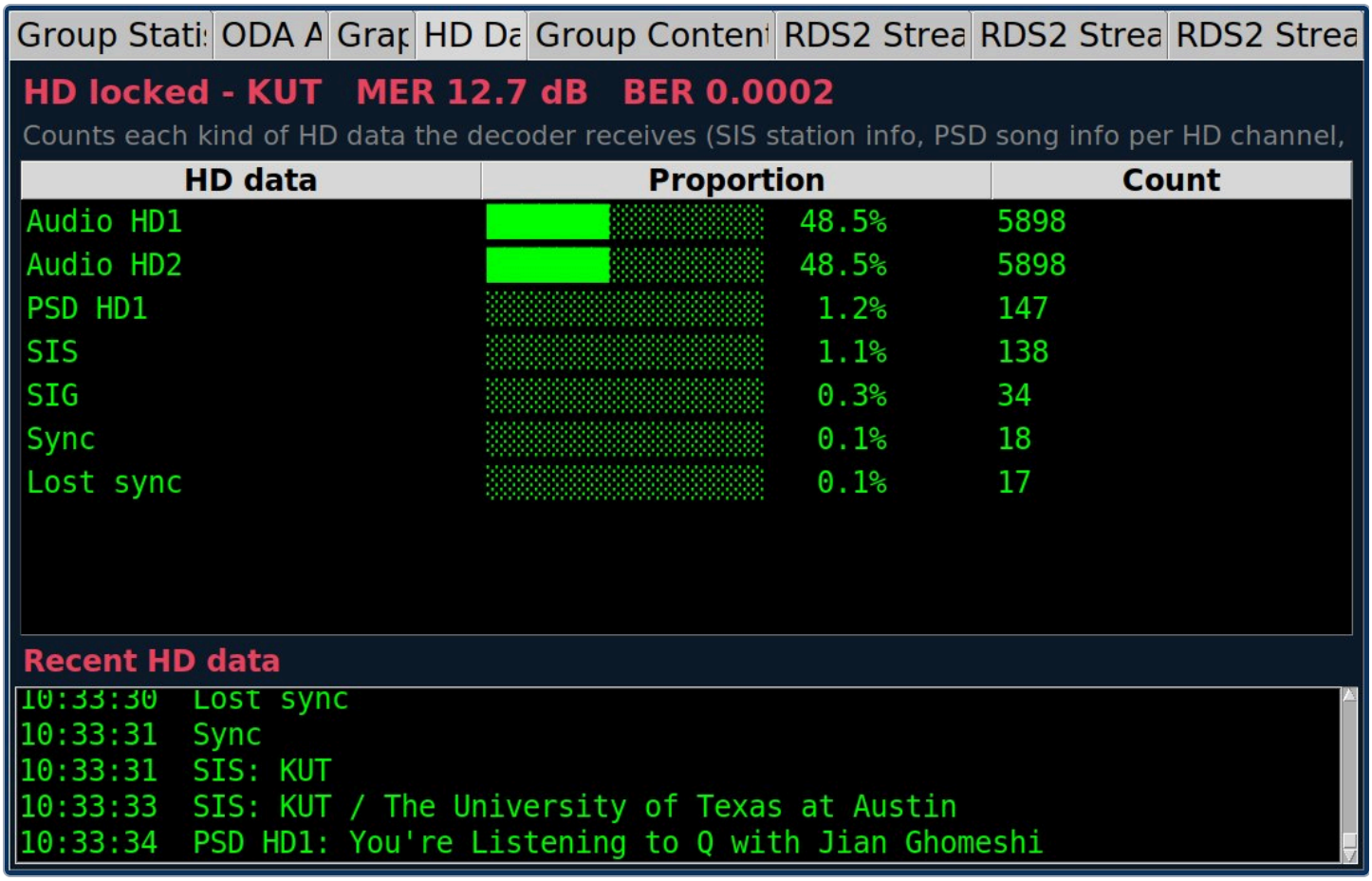
View › Open Group Analyzer Window: Group Statistics (0A–15B counts), ODA AID (applications such as RT+ and TMC), Graph, Group Content HEX, RDS2 Stream 1–3, and **HD Data**.

Weak stations and ODAs: the Monitor tries to decode every group it receives, whether or not a 3A group announced it, so a weak station you only hear for a few seconds still gives you whatever gets

through (RT+, TMC and the rest) even if you never catch its 3A. Only the standard, registered ODAs are decoded. On a noisy signal a damaged group can look like a 3A announcement of an AID that doesn't exist (often two RadioText letters, such as 0x414E = "AN"). The ODA AID tab shows these as they arrive, but DX Mapper doesn't list them.

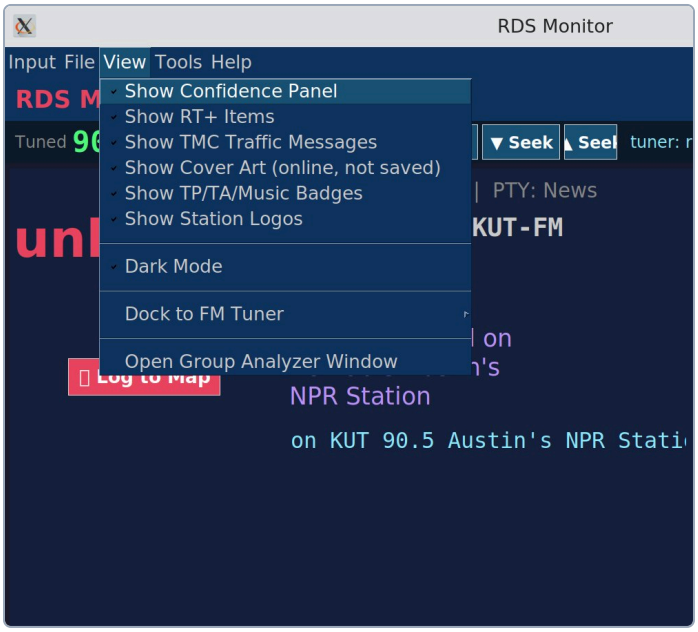


Group Statistics.



HD Data tab: counts of each kind of HD data, then a log of recent HD data (KUT recording).

5.9 View and Tools menus

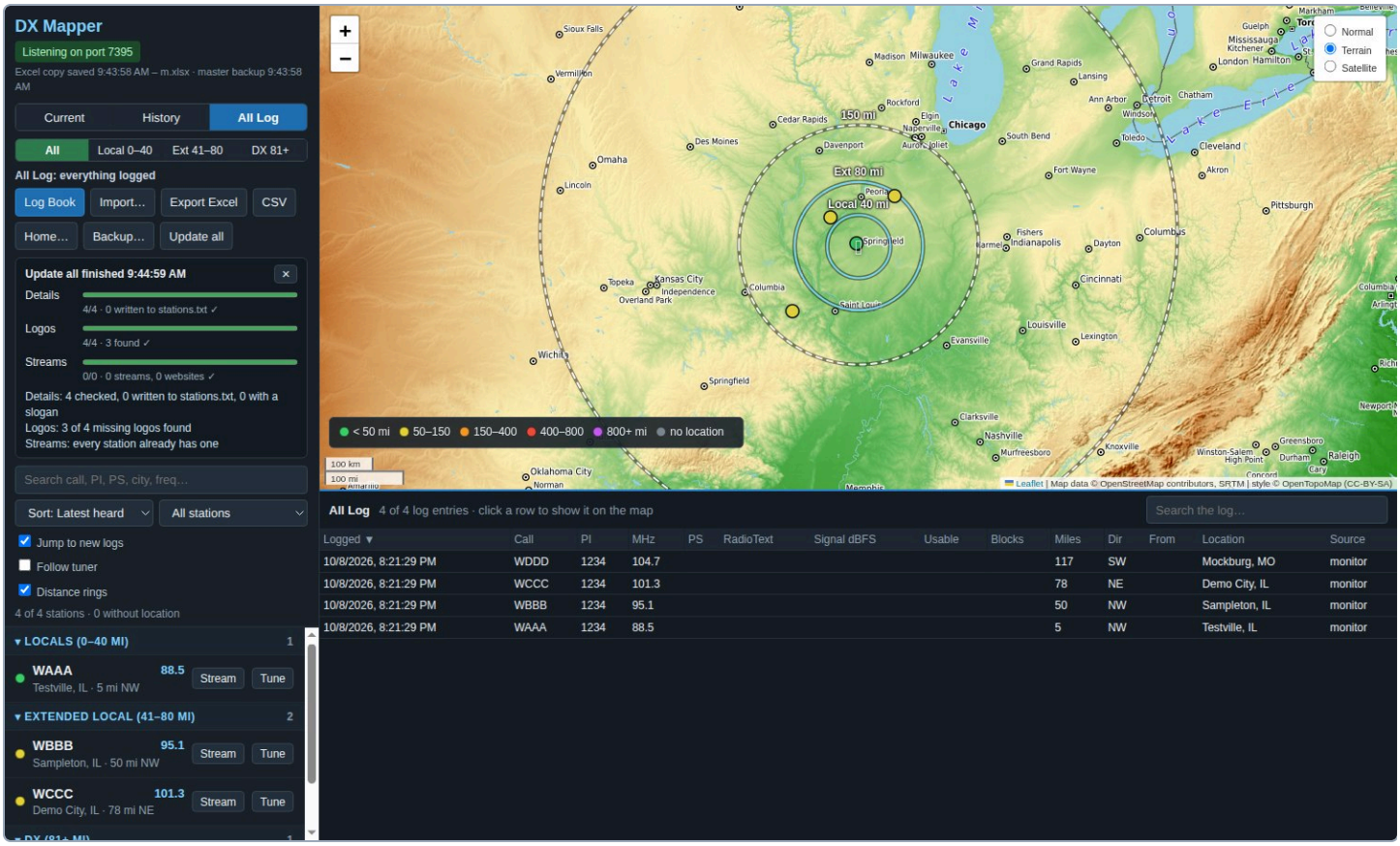


View menu.

- **View:** show/hide the confidence panel, RT+, TMC, cover art, TP/TA badges and logos; Dark Mode; Dock to FM Tuner; Group Analyzer.
- **Tools:** Log Current Station to DX Mapper, Auto-log, Set Home / Away Location, Use Home / Away location, GPS Location, Manual Logo Lookup, Download Missing Logos, reset the station counters.

- **File › Generate Report** (Current Station or All Stations) saves a full report (into the `reports` folder by default).

6. DX Mapper



DX Mapper with made-up test stations: sidebar on the left with the Update all summary, map on the right.

6.1 The sidebar

- **Listening on port 7375**: ready for logs from the RDS Monitor. **Now hearing** shows what the tuner is on right now (a pulsing pink marker on the map).
- **Current** (logged in the last 1–24 hours), **History** (all time, today, last 7/30 days, picked dates, or a day you click) and **All Log** (a table of every catch under the map).
- **All / Local 0–40 / Ext 41–80 / DX 81+** narrows the markers, list and table.
- Buttons: **Log Book**, **Add station…**, **Import…**, **Export Excel**, **CSV**, **Home…**, **Backup…**, **Conditions…**, **Screenshot**, **Update all**.
- **Add station…** logs a station that sends no RDS. It reads the frequency the FM Tuner is on (or type one) and lists every FCC station on it, closest first, with city, type (FM, LPFM, translator), power and distance. Mostly for IDing locals, but DX stations are listed too. Near the edge of coverage two stations can share a frequency, so nothing is picked for you: click **Add** on the one you hear. Type to narrow the list: call letters, a slogan, "City, ST", a state ("ky" or "Kentucky") or a region ("central ky"). A station you added stays as you entered it: RDS of a different station on that frequency becomes its own entry, and RDS of the same station only fills in blanks such as the PI.
- **Now hearing with no RDS:** when the Tuner is on a frequency with no RDS, the Now hearing bar shows the station you logged there, marked "(no RDS)", with the map pulse, coverage outline and Follow tuner. If two or more are logged on that frequency it shows a button for each so you pick the one you hear (it never guesses; **change** picks again). If none is logged it shows an **Add station…** button. RDS always takes over when it decodes, and nothing is logged by itself.

- **Search**, **Sort**, **Follow tuner** (the map follows whatever you tune), **Local / Ext rings** and **100–400 mi rings** (each turns on and off by itself and is remembered).
- **Auto-center map** (on by default): picking, logging or tuning a station zooms the map so the station and your location both fit. Untick it when you want the map to stay put.
- **Coverage when tuned**
- **County / state lines** (on by default): county lines (thin, dashed yellow) and state lines (white) on all three maps. They come with DX Mapper, so they work without internet.
- **Weather radar** (off by default): live US Doppler radar (NWS NEXRAD, from the Iowa Environmental Mesonet) over the map, refreshed every 5 minutes. Use it to see storms near you or a transmitter; warm air ahead of a front can bring tropo, and summer storms can go with E-skip.
- **Hide my location** (remembered): for sharing screenshots of your catches. It hides your house (or car), the Local/Ext and 100–400 mi rings, the lines from home and every distance and direction on screen, because a transmitter plus its distance and direction would pinpoint your home. Instead you get one Local-size circle shifted 10–20 miles off-center, so people see your general area but its middle isn't your house. Untick it and everything comes back; your log keeps all the distances.
- **Screenshot** button: turns on Hide my location, saves a picture of the DX Mapper window in *settings*|*Screenshots* (the folder opens so you can post it), then puts the checkbox back how you had it.
- **Conditions…** button: opens free skip-condition websites in your browser – Hepburn's tropo forecast, DXMaps live E-skip contacts and PropQuest's E-skip MUF. DX Mapper can't predict skip itself; these sites are the best free guide.

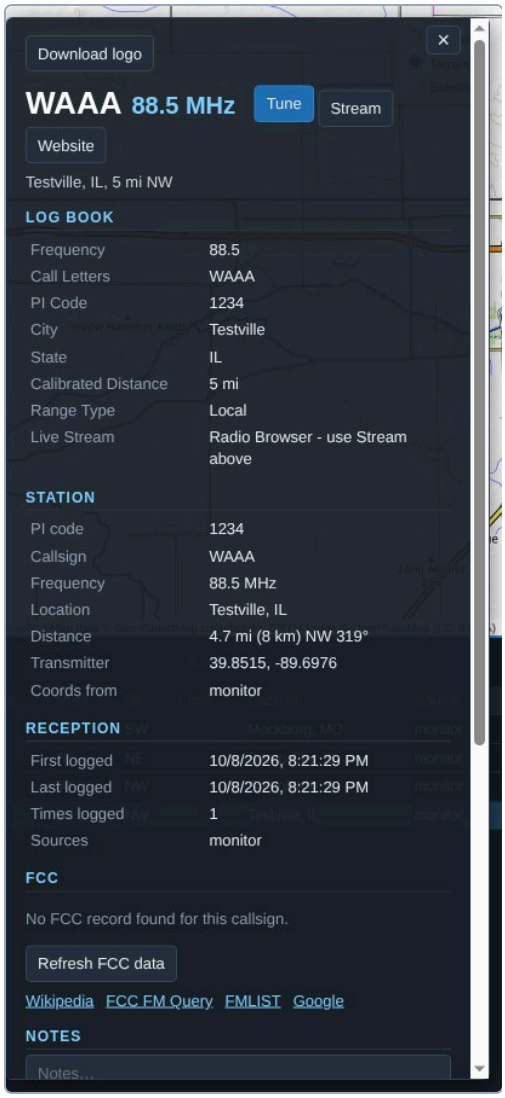
  : draws the FCC service contour (pink outline) of the station you're tuned to – the area the FCC expects it to cover, worked out from its power, antenna height and terrain (60 dBu, or 54/57 dBu for class B/B1). Outlines are saved in *settings*|*Coverage* for 30 days. Real reception often goes well past it.
- **Station list:** Locals and Extended local in frequency order, then DX. Each row has **Tune**, and **Stream** when a stream was found.

6.2 The map

- **Normal**, **Terrain** or **Satellite** (top right). Tiles you've seen are saved, so they also work offline.
- Marker colour = distance band. **Local / Ext rings**: solid blue rings at your Local and Extended local distances (40 and 80 miles unless changed under Home…). **100–400 mi rings**: white rings solid at 100, 200, 300 and 400, dashed at 150, 250 and 350.
- Picking a station keeps your tuner location (home or away) in the middle of the map, draws a line out to the transmitter and zooms so both fit beside the details panel.
- Translators are mapped at the translator's own tower, with “Relays WXXX” in the details.
- HD1 is always the analog station's own entry. HD2–HD8 get their own entries (WXYZ-HD2…).

6.3 Station details

Click a marker or a station in the list.



- **Logo** (or **Download logo**), call letters, MHz and slogan. An HD2–HD8 entry looks up its own logo from the channel's website listed on Wikipedia and otherwise shows the main station's. Every channel's logo is saved in the RDS Monitor's logos folder (WXYZ.png, WXYZ-HD2.png…); a logo decoded off the HD signal replaces the online one, and online lookups never replace a logo you already have. A station with no logo of its own (e.g. a King of Kings Radio station) gets its network's logo. Facebook's grey "no photo" person and Wikipedia's studio/tower photos are skipped. **Change logo** shows every logo found online (Wikipedia, Facebook, the network, the station's website) as small pictures: click one, or upload your own picture. Your pick replaces the old logo right away, and nothing automatic (not the online lookup, not an HD logo) changes it back. A **translator** (e.g. W293BK relaying WTCO) shows its own call letters but uses the logo of the station it relays (WTCO.png); no separate logo is looked up for it. The parent can be an AM station (e.g. WTCO 1450 AM and its FM translator W294CO): DX Mapper finds it from the FCC record, WTFDA's *Relay of*, the station's Wikipedia page or what you type in **Fed by**. An **HD2-HD8 channel that simulcasts an AM station** (Wikipedia lists it as e.g. "HD2: Simulcast of WTCO 1450 AM") keeps its own call (WXXX-HD2) and uses the AM's logo; if Wikipedia doesn't say so, type the AM's call in that HD channel's **Fed by**.
- **HD Channels**: on a station with HD, the Log Book HD column says 1 and this line lists its channels (HD1, HD2 · name…). Click one to open that entry; each HD entry links back. Channels Wikipedia lists that you haven't logged show without a link.
- **Now playing bar**: while the open station is the one the RDS Monitor is decoding, a scrolling bar under the call letters shows the song or RadioText (hidden when it's not on the air).
- **Sister stations & same market** (just above Edit): sister stations from Wikipedia's “Sister stations” line plus stations in your log with the same owner, and the FM stations the FCC lists for the station's city plus stations in your log from the same city and state (your Log Book City/State first). Stations in your log are links; the rest are plain text. Saved for a month.
- **Translators**: a translator shows **Fed by** (the station or HD channel it relays, e.g. WXYZ-HD2) and the parent lists its translators, both clickable. It comes from the FCC and the station's Wikipedia page; if it's wrong or missing, type it in *Edit › Fed by*.
- **Tune** tunes the FM Tuner to the station, off the air.
- **Stream** opens the station's internet stream in your web browser.
- **Website** opens the station's own website.
- **Log Book** section: your spreadsheet's columns for this station.
- **Full report** opens the latest complete RDS Monitor report as plain text.
- **Analog** section (titled WXYZ): the **Decoded RDS** list. **HD** section (titled WXYZ-HD, or WXYZ-HD2 on an HD2 entry): best MER, last HD lock, HD channels and the **Decoded HD** list. Each list shows

every item that can be decoded, with a green ✓ once it was observed and a grey – until then; the actual decodes are in Full report. Only defined (registered) ODAs are listed: one the station announced that isn't already on the list, such as iTunes Tagging, gets its own ✓ line. Unregistered AIDs, usually a damaged group on a weak signal, are left out here and in the Modern Log Book panel. **Location**: city, distance and bearing, transmitter. Click a heading to fold that section away; it stays folded.

- **Reception**: first/last logged, times logged; **FCC**: licensee, power, height; every log with its report.
- **Edit**: call letters, MHz, coordinates, slogan. New call letters look up their own location, and the RDS Monitor shows the change within a second. A slogan you type always wins over Wikipedia.
- **WTFDA database** section: what *db.wtfda.org* lists for the station (format, slogan, PI, PS, RadioText, ERP/HAAT, remarks). Every station is looked up there once, in the background, 30 seconds apart as the site asks. A catch with only a PI code is looked up by PI and gets its call letters when one station on that frequency has it. WTFDA only fills blanks, and it ranks above Wikipedia but below your own decodes, your edits and the FCC transmitter site. Its RadioText is replaced as soon as the station's own decodes.

6.4 Update all

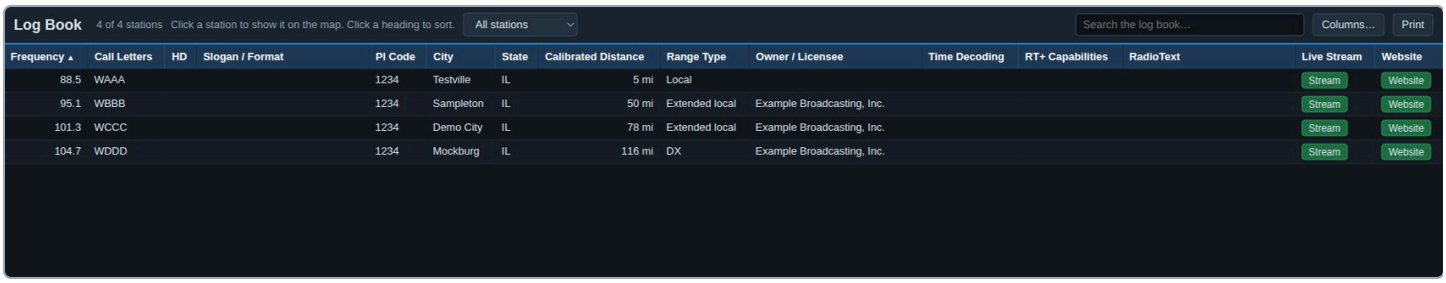
One button that goes through your whole log in the background:

1. **Details:** fills in call, city, coordinates and slogan for the RDS Monitor (its background stations.txt). Lines you typed are kept.
2. **Logos:** downloads a logo for every station without one. Logos you have are kept.
3. **Streams:** looks up a live stream and the station's website for every station missing one. This runs alongside the other two.

A box under the buttons shows a progress bar for each step and, when it's done, a summary that stays until you close it with ×. Anything that went wrong, such as no internet, shows in red. A lookup that couldn't reach the internet isn't saved, so it is tried again next time.

Where streams come from, in order: Radio Browser, iHeartRadio, TuneIn, the station's Wikipedia page (its Webcast line), and last the Listen Live link on the station's own website. Websites come from Radio Browser, iHeart or Wikipedia's Website line, never guessed from the call letters, because many stations use their slogan in their address. A station only gets a stream when its call letters (and frequency/HD channel, when listed) match. Not every station has one. New catches are looked up automatically.

6.5 Log Book



Log Book with made-up test stations.

- One row per station, with your spreadsheet's columns: Frequency, Call Letters, HD, Slogan / Format, PI Code, City, State, Calibrated Distance, Range Type, Owner / Licensee, Time Decoding,

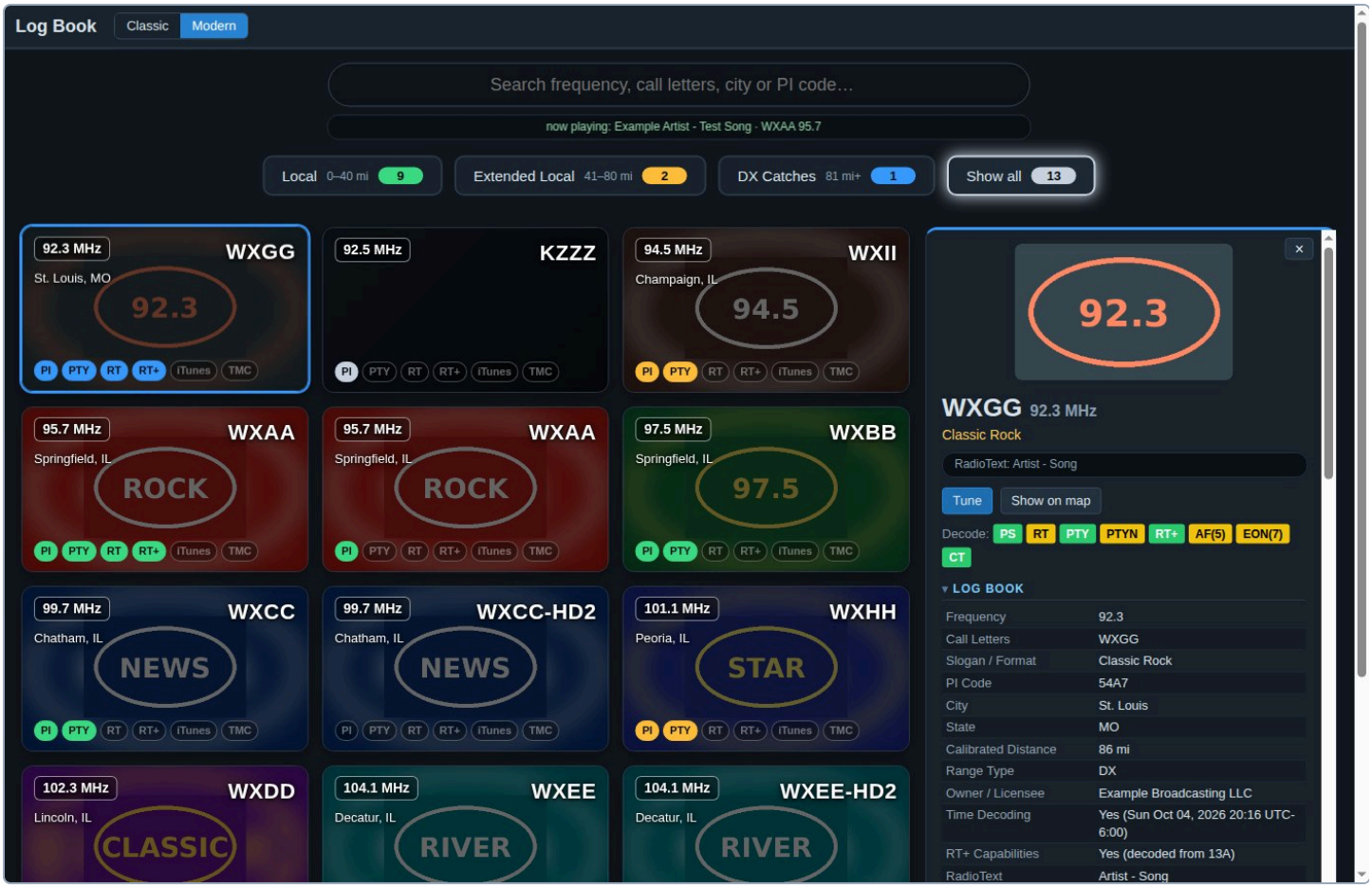
RT+ Capabilities, RadioText, **Live Stream** and **Website**.

- Click a heading to sort; search any column (“ky” finds Kentucky and “kentucky” finds KY; “campbellsville ky” works too); filter All / Locals / Extended locals / DX / HD; **Print**.
- Click a station and the map jumps to it.
- **Editing, like Excel:** click a cell to pick it; drag, Shift-click or Ctrl-click to pick several. Start typing (or double-click / F2), then press Enter: what you typed goes into every picked cell. **Delete** clears every picked cell; Esc cancels. **Undo my edits** puts the picked cells back to what DX Mapper fills in. Delete never removes a station; that's still the Delete button in Station Details. Call letters can't be blanked and MHz must be a number.
- **Source** shows which dongle made the catches, newest first; the tuner sends it with each log. **Notes** is yours: type reception notes like “crystal clear” or “faded in and out”. Import never fills it and Clean up never deletes it.
- **Clean up** tidies RadioText and deletes the old imported notes and any imported columns that aren't Log Book columns. A progress bar shows while it runs.

  ⚠ **Warning:** Clean up **permanently deletes** that data and Undo my edits can't bring it back. It asks you to confirm first, then saves a copy of your whole log as **settings\Backups\Before Clean up** *date*.**db** before deleting anything. To undo it, open **Backup…**, choose that file and restore it. Your Log Book columns and your own Notes column are never touched.

- **Splatter doubles:** a strong station can show up on other frequencies when conditions are right. When an **auto log** catches a PI (or call letters) that's already in your log on another frequency, it's added to that station instead of making a new one, unless the FCC lists a translator for it on the new frequency. Pressing **Log** in the RDS Monitor yourself always logs it where you're tuned. If the FCC says the new frequency is the station's real one, the station moves there.
- **One entry per station:** when a station ends up with the same call letters and frequency as another entry (an FCC fix, a catch or your own edit moves it there), DX Mapper merges the two right away.
- **Duplicates…** lists stations listed more than once: the same call letters or PI twice on one frequency (say, after typing call letters into a row), or again on another frequency, keeping the FCC frequency (else the one logged most). Untick any you want to keep apart, then **Merge ticked**: every log moves into the kept station. A copy of the log is saved first as **settings\Backups\Before Duplicates** *date*.**db**. Translators already logged as translators are left alone.
- **RadioText** keeps the station's pattern, like “Song by Artist on The Wave”: the station's own words with the song playing replaced by *Song* and *Artist*, so it helps confirm which station you heard. Promos, sports, church programs, ads and phone/web lines are never saved there. RadioText you type, import or already had is cleaned the same way: a real song becomes the pattern, a cell of jumbled captures is boiled down to one line, and with no song the slogan or ID line is kept.
- **Decoded RDS columns** (PI, PS, PTY, RT, RT+, PTYN, Long PS, AF, EON, TMC, CT, TP, TA, DI) are the same checklist as Station Details: a ✓ means one of your RDS reports decoded it. Click a heading, say **TMC**, to bring every station you've decoded TMC on to the top, or type **TMC** in the search box to show only those. They fill in by themselves and can't be typed in.
- **Columns…** picks which columns show and print. Everything starts on; your choice is saved in the settings folder. Excel export always keeps every column, including the stream and website addresses.

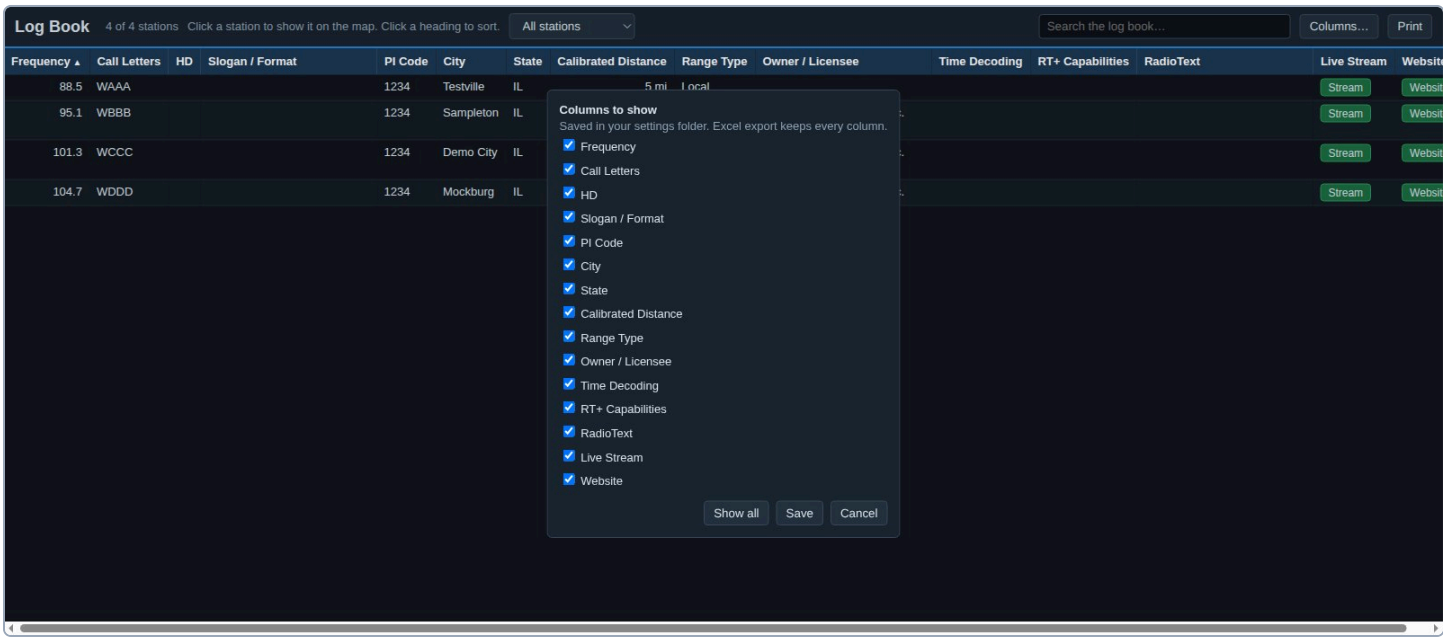
6.5.1 Modern view



Modern view with made-up test stations, one picked.

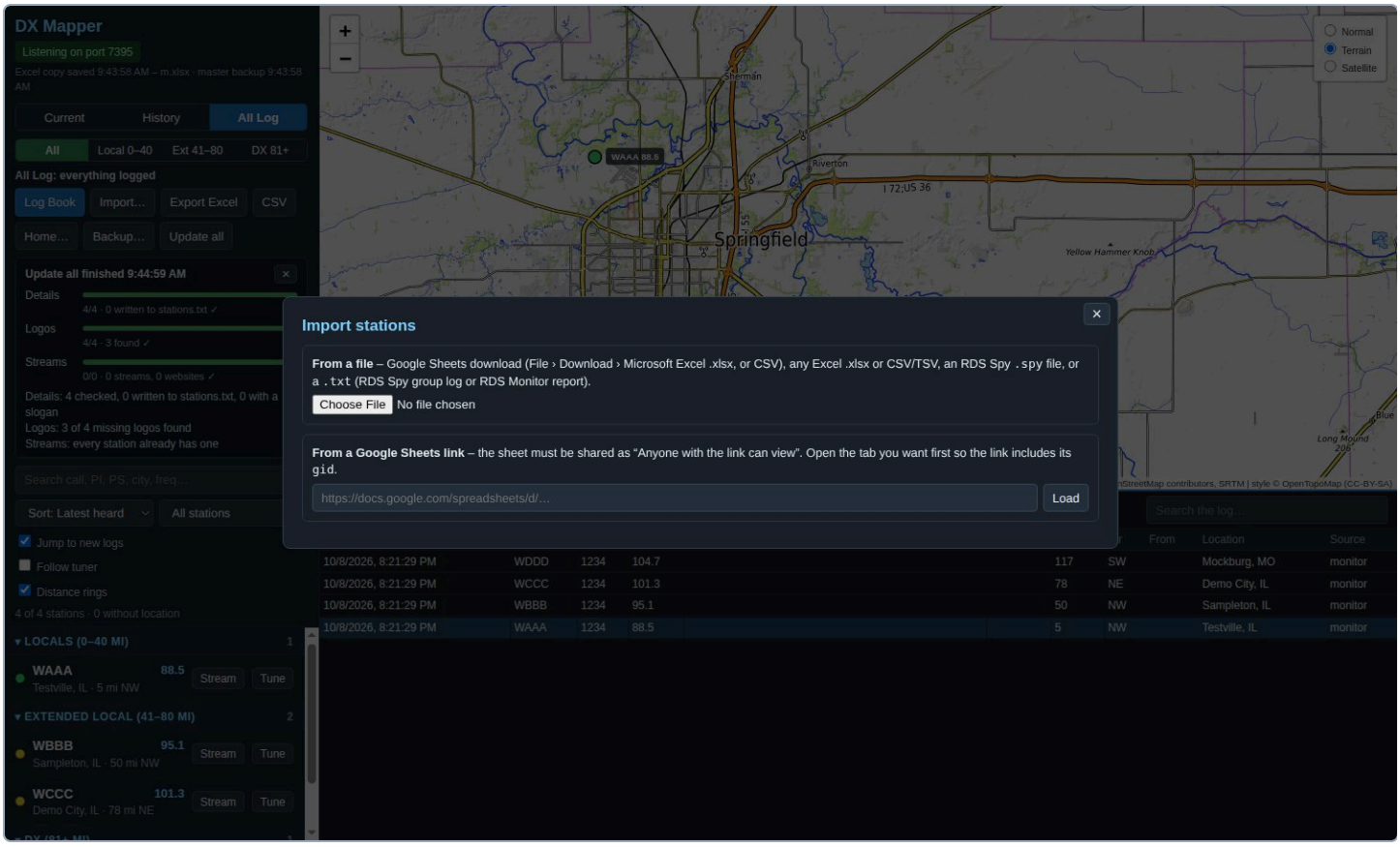
- The **Classic | Modern** switch next to the Log Book title flips between the spreadsheet and a card view. DX Mapper remembers which one you used last. Everything in this section is the Modern view; Classic is unchanged.
- **Search** (the big box) matches frequency, call letters, city, state, PI code or slogan as you type. It forgives a little: “wxz” finds WXYZ and “sprinfield” finds Springfield. Under the tabs it tells you if other tabs have matches too.
- The thin bar under it shows what the RDS Monitor is decoding now (“now playing: Artist - Song”).
- Tabs: **Local** (green), **Extended Local** (amber) and **DX Catches** (blue), each with its count, and **Show all**, every station in the Log Book as cards (each card keeps its green, amber or blue border; stations with no distance yet show grey). The miles follow your Local / Extended local settings and the Range Type column. Stations with no distance yet show only in Show all and Classic.
- Each card shows frequency, call letters and city, with the station's logo behind a dark shade. The **PI PTY RT RT+ iTunes TMC** lights turn on once you've decoded that item for the station. **iTunes** lights when the station announces iTunes Tagging (ODA C3B0).
- **Call letters:** a regular station is just its call letters, like **WAPD**, and HD1 is that same station. HD2–HD8 get the ending, like **WAPD-HD2**, each on its own card (also when only your sheet's HD column says HD2). So two cards with the same call letters on one frequency are a real double; **Duplicates…** in Classic merges it.
- Click a card to open the station panel on the right (× or Esc closes it): logo, slogan, live song when you're tuned to it, **Tune**, **Show on map**, and **Stream** / **Website** when the station has them.

- Below that come the same sections as the map's Station Details: the **Decode** badges, **Log book**, **Full report**, **Analog** (Decoded RDS checklist), **HD Radio**, **Location**, **Reception**, **FCC** (ERP, HAAT, licensee, links), **WTFDA database**, **Translators**, then **History** (the first paragraph of the station's Wikipedia article, saved in **settings\wiki_summaries.json** and checked again after 30 days) and **Log history** with each log's Report. Click a section heading to fold it; folds are shared with the map's panel. HD and translator links open that station's card.
- **Edit info** at the bottom has every Log Book column you can type in (Frequency through Notes; the RDS ticks come from your reports). Change any of them, then **Save** (or Enter). It's the same as typing in the Classic cells, so the Excel copy updates and the card refreshes right away.
- **Delete station** under Edit info deletes the station and all its logs, and the ✕ in Log history deletes one log entry. Each one opens a confirmation popup first (**Cancel** or Esc keeps it). Deleting can't be undone except by restoring a **Backup…**.
- **It follows the map:** whatever station DX Mapper opens (a click on the map or list, **Jump to new logs**, **Follow tuner**) opens in the panel too, on the right tab. Those two map checkboxes decide what follows; the Log Book has no switch of its own.
- **Open station card** in the map's Station Details opens that station's card here, in the Modern view. If no Log Book is open, one opens in your web browser. Going the other way, **Show on map** in the panel moves the map to the station.



Columns… dialog.

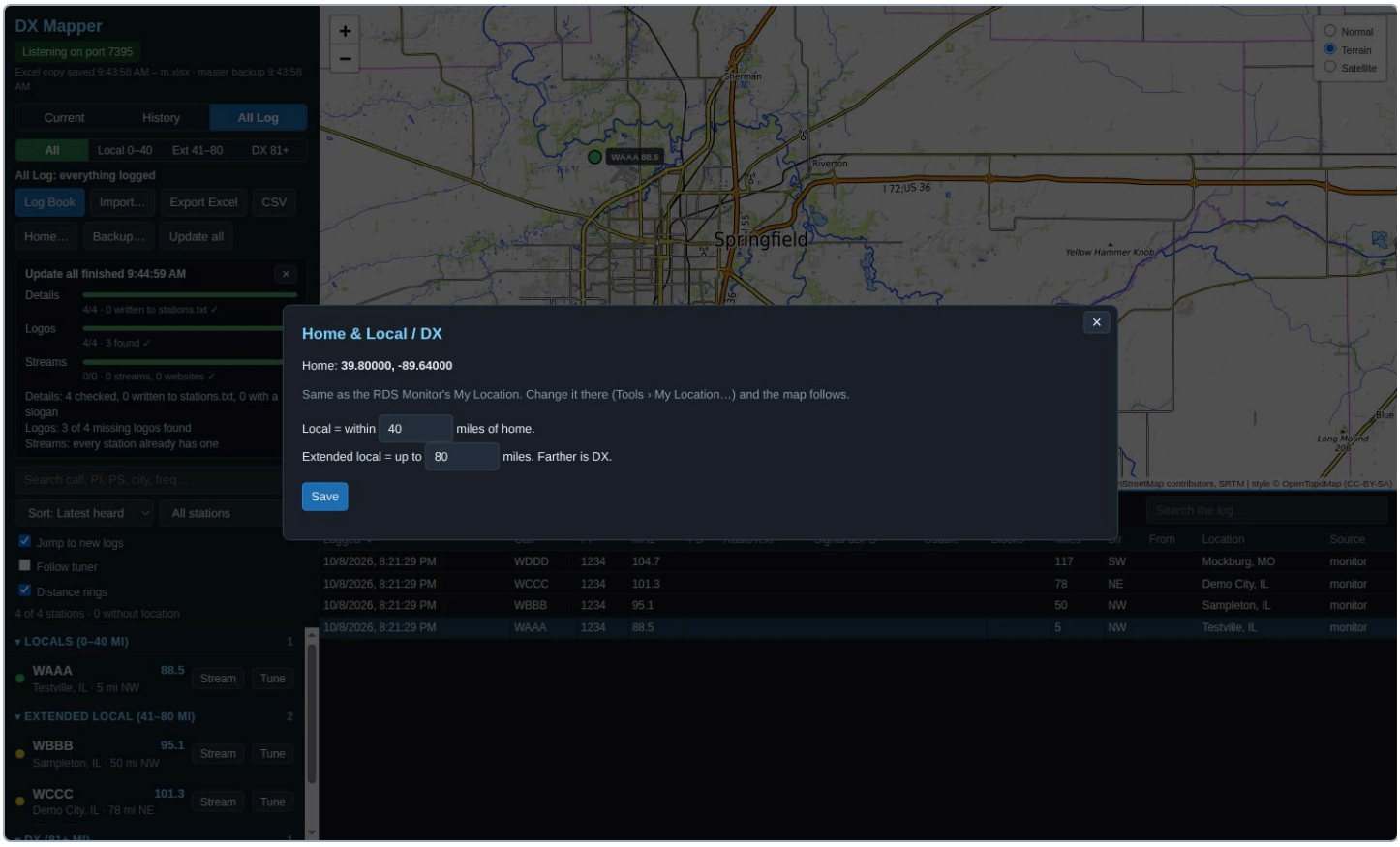
6.6 Importing old logs



Import stations.

- **Google Sheets:** paste the link (shared as "Anyone with the link: Viewer"), or download it as Excel/CSV and pick the file. Every tab with log headings is read.
- **Excel .xlsx**, **CSV**, RDS Spy **.spy** and .txt group logs, and RDS Monitor reports.
- Columns are recognised automatically; your log book columns go into each station. Notes and any other columns are not imported; **Clean up** in the Log Book deletes ones an older version imported.
- Imports run in the background while you keep using the map. Importing again merges into the existing stations.
- **An import only fills blanks.** What your log already has (your own decodes and edits) is never replaced by a sheet.
- HD rows (`WKQX-HD2`, `101.1 HD2` or an HD column) become their own stations and keep exactly what your sheet says.

6.7 Home and range miles



Home… dialog.

Home follows the RDS Monitor's active Home or Away location. Here you set the Local and Extended local miles (40 and 80 by default).

6.8 Several logs (Historical, Current, trips)

Each log is its **own file**, so a problem with one never touches the others. Your first log is `settings\dx_log.db` (called **Main** until you rename it); every other log is `settings\Logs\<name>.db` with its own Excel copy and its own backups in `settings\Backups\<name>`.

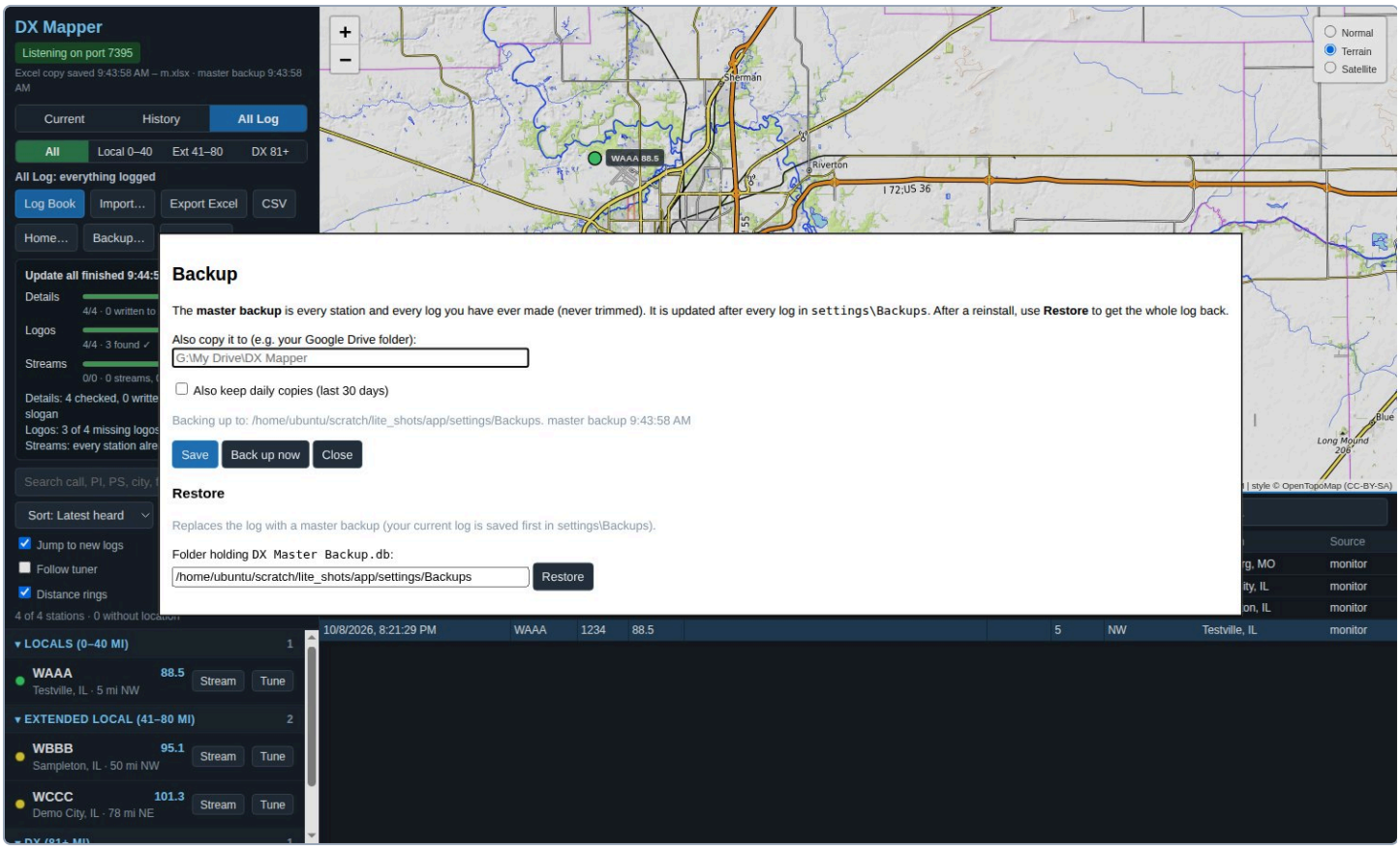
- **Logs…** (under the feed line, top left) shows **Log: Current + Historical**. It opens the list of logs with **New log**, **Rename…** and **Delete…**.
- **Tick the logs to show together.** The **first one you tick** (marked ✎) gets every new catch (RDS Monitor Log button and Auto-log), every edit and every import. The others are **view only**. To change which one is first, untick them all and tick them again in the order you want, then press **Show ticked**.
- **One pin per station.** A station that is in two ticked logs gets one pin and one card. Its info comes from the first log; anything missing there (slogan, RadioText, notes, logo…) is filled in from the other log. The card says *Also in Historical (12 logs, first heard 3/2019)*, and Log history lists both logs' entries, each tagged with its log name and showing where it was heard and how far.
- **Station only in a view-only log:** it still shows on the map and in the Log Book (gray italic rows), with a badge of its log's name. Its card says *From your Historical log – view only*; Edit and Delete are hidden.
- **Copy to Current:** on a view-only station's card, copies the station into the first log. Already there? Only its blanks get filled. Each view-only log entry also has a **→ Current** button that copies that

catch (with where it was heard). Copying any HD channel, or any of its catches, copies the **whole HD family** (the main station and its HD2–HD8) with their entries from the same catch. The view-only log is never changed. Copying the same catch twice doesn't double it.

- **Also log new catches into the other ticked logs** (in Logs…) is **off by default**. With it on, every new catch goes into all the ticked logs. Ticking it opens a warning popup so nobody turns it on by accident: only **Turn it on** turns it on; **Keep it off**, closing the popup or Esc leave it off. Edits, deletes and imports still go only into the first log.
- **Where you heard it:** every log entry keeps the spot it was heard from (Home, Away, GPS), so catches from an old home keep that spot in any log. Distances on the map and list are from where you are now.
- **Delete…** asks twice, then removes just that log: its file, Excel copy and its backups in settings\Backups. A copy in your Google Drive backup folder is kept. The log taking catches can't be deleted or renamed; tick another one first.

> Example: rename Main to **Historical** (the catches from your old home), make **Current** for where you live now, then tick Current first and Historical second. You hunt with the old info on the map, and new catches go into Current only.

6.9 Backup and restore



Backup… dialog.

- The **master backup** holds every station and every log ever made, never trimmed, in `settings\Backups\DX Master Backup.db` (plus an .xlsx copy). It is refreshed after every log and import. Each other log has its own master backup in `settings\Backups\<log name>` (and a <log name> folder in your Drive folder).

- Put your Google Drive folder (e.g. `G:\My Drive\DX Mapper`) in **Also copy it to**, so a copy lives off the PC.
- **Daily copies** (off by default) keeps one dated copy per day for 30 days.
- **Restore** brings the whole log back after a reinstall: point it at the folder holding `DX Master Backup.db`.
- `settings\DX Log.xlsx` is an Excel copy rewritten after every change; **Export Excel** makes a dated copy any time.

> **DX Log.xlsx is a copy for viewing. Edits typed into it in Excel are NOT read back and get overwritten. Make all changes in DX Mapper (Edit info or the Log Book). To bring in an edited sheet, use Import… (Excel or Google Sheets); it only fills blanks.** The same bold warning is on the top row of every sheet.

7. Your data – the settings folder

Everything you edit, and everything the three programs save for you, lives in the one **settings** folder next to Start Radio. Updates never touch it and the ZIPs never contain it.

Program	Its files in settings
FM Tuner	tuner_presets.json (presets), tuner_layout.json (panels, size, layout presets), hd\ (HD decoder)
RDS Monitor	stations.txt (background fallback), logos\, monitor_settings.json (Home/Away), station_count.txt
DX Mapper	dx_log.db (your first log), Logs\ (your other logs, each with its .xlsx), DX Log.xlsx (Excel copy), Backups\, logger_settings.json (map, miles, hidden Log Book columns), TileCache\, dx_mapper.log
Arrange Windows	arrange.json (which screen each program goes on)

> Logo names: `WKQX.png` is the station's logo, used for analog and HD1. HD2 and up have their own: `WKQX-HD2.png`, `WKQX-HD3.png` …

8. Troubleshooting

Problem	What to do
“'python' is not recognized” / “'py' is not recognized”	Re-run the python.org installer, choose Modify, tick “Add Python to environment variables” and “py launcher”.
No window / tkinter error	Reinstall Python with “tcl/tk and IDLE” ticked.
“rtlsdr.dll not found”	Copy the x64 rtlsdr.dll and libusb-1.0.dll into the main folder (chapter 2).
“No RTL-SDR device found”	Run Zadig for this USB port; replug the dongle; close other SDR programs (SDR#, etc.).
Bias-T “not supported”	Use the RTL-SDR Blog rtlsdr.dll.
No sound	Pick your speakers in Output device, press Refresh, make sure Audio is ticked.
Remote page won't open on the phone	Check the phone is on the same Wi-Fi as the tuner PC (not mobile data), and that Allow remotes is ticked. If Windows' firewall question was cancelled: Start › “Allow an app through Windows Firewall” › Change settings › tick Private next to every Python entry.
Remote page shows but no sound	Press Listen on this device. The tuner's audio must be running (Audio ticked, receiver started). Turn up the phone's media volume.
Remote shows PS instead of call letters / no logo	Start the RDS Monitor and DX Mapper (Start Radio does); the call, slogan, city and logo come from them.
Strong locals smear across the band / OVERLOAD	Lower the gain (start at 0 dB), lower the LNA gain, or add an FM notch.
RDS Monitor says connecting…	Start the FM Tuner first; allow Python through Windows Firewall for this PC.
Station shows “unknown” or the wrong call	Fix the station's call or PI in DX Mapper (Edit info or the Log Book); the monitor picks it up within a second.
Log to Map says Map not running	Start DX Mapper (or Start Radio).
HD: no lock on this frequency	The decoder is installed; the station has no HD or it's too weak. Look for HD shelves on the waterfall (4.6).
HD decoder not installed	Audio panel › Add HD decoder… and pick the decoder ZIP.
Update all found no streams	Read the red lines in the Update all box. No internet or a blocked connection is reported there and retried next time.
Map tiles don't load	Needs the internet the first time a place is viewed; places already seen work offline.

Program closed with an error	DX Mapper writes settings\dx_mapper.log. Send it, or a screenshot, when asking for help.

9. Glossary

Term	Meaning
PI	Program Identification: a 4-digit code every RDS station sends; in the US it encodes the call letters.
PS	Program Service name: the 8-character name on a car radio display.
RT / RT+	RadioText: up to 64 characters of text. RT+ tags parts of it (title, artist…).
PTY / PTYN	Program type (News, Rock…) and its custom name.
AF / EON	Alternative Frequencies and Enhanced Other Networks: lists of other frequencies and stations.
TMC	Traffic Message Channel.
ODA	Open Data Application: extra RDS services such as RT+ and TMC, announced in 3A groups.
Group	One 104-bit RDS message (0A, 2A…).
MPX	The composite signal after FM demodulation: audio, pilot, stereo, RDS and subcarriers.
SCA	Subsidiary Communications Authorization: extra analog audio at 67 or 92 kHz.
HD Radio / IBOC	Digital audio sent alongside the analog signal, in sidebands either side of the station.
MER / BER	HD signal quality: Modulation Error Ratio (higher is better; green 10 dB+ good, yellow 6–10 fair, red under 6 weak) and Bit Error Rate (lower is better; under 0.01 is solid). The HD Data heading takes the MER colour.
Translator	A low-power relay of another station on a different frequency.
Tropo / Es	Tropospheric ducting and Sporadic-E: conditions that bring in distant FM stations.
Bias-T	DC power sent up the antenna cable to an LNA or active antenna.

10. For experimenting

The parts talk to each other only on this PC (127.0.0.1); only the remote page (7377) is open to your home network. Each folder's README.md lists every option.

Port	From → to	What
23	FM Tuner → RDS Spy and other RDS programs	RDS groups (ASCII G)
7376	FM Tuner → RDS Monitor	Same RDS groups plus RDS2
7373	FM Tuner → RDS Monitor	Telemetry: frequency, signal, stereo, HD
7374	anything → FM Tuner	Commands, e.g. TUNE 101.1
7375	RDS Monitor → DX Mapper	Logs and Now hearing
7380	DX Mapper	The map window
7377	phones / other PCs → FM Tuner	Remote page and audio (home network, only when switched on)

Screenshots use test recordings and made-up test stations, not live reception. HD Radio has so far been tested only with nrsc5's recorded KUT sample, not off the air.

11. Credits

DXMapper Suite stands on the work of many people and organizations. Full details and license texts are in **THIRD_PARTY_NOTICES.txt** (in the public copy, *DXMapper Suite v<release>.zip*).

Used for	Credit
Map	Leaflet (BSD 2-Clause) – Volodymyr Agafonkin and CloudMade
Map tiles	© OpenStreetMap contributors (ODbL); © Esri and its data providers (USGS, NASA, CGIAR and others); imagery © Esri, Maxar, Earthstar Geographics
Weather radar	Iowa Environmental Mesonet, Iowa State University (NWS NEXRAD)
Place search	Nominatim, © OpenStreetMap contributors
County / state lines	us-atlas and topojson-client (ISC) – Michael Bostock; U.S. Census Bureau data
Map screenshots	html-to-image (MIT)
Remote page QR code	Segno (BSD 3-Clause) – Lars Heuer
Station data	FCC; WTFDA FM database (Worldwide TV-FM DX Association); Wikipedia (CC BY-SA) and Wikimedia Commons
Streams, logos, album art	Radio Browser, iHeart, TuneIn, Apple iTunes Search, Facebook page pictures
Standards	NRSC-4-B and NRSC-5 (National Radio Systems Committee); IEC 62106 / EN 50067 (RDS); RDS Forum (RT+, ODA list); ISO 14819 (TMC ALERT-C)
Compatible with	RDS Spy (Jan Kolář) and SDR# RDSOutput (Airspy)
Installed separately	Python, numpy, sounddevice, scipy, requests, Pillow, pyserial, pywebview; RTL-SDR driver, Zadig, SoapySDR, nrsc5

HD Radio is a trademark of Xperi Inc. All other names, call letters and logos belong to their owners. This project is not affiliated with or endorsed by any of them.